

METHODS

A Novel Fourier-Based Thresholding Method for Efficient Drone Detection in Images

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ABSTRACT Drone detection in images is a critical task for security and surveillance applications. Traditional thresholding methods are commonly used for contrast-based object detection but often suffer from high false positives, low precision, or computational inefficiency. In this paper, we propose a novel Fourier-based adaptive thresholding algorithm that utilizes Fourier transform to approximate the image histogram, enabling robust identification of multi-modal background distributions for precise threshold selection. The method introduces several key innovations: 1) low-pass filtering of the histogram via FFT to suppress noise while preserving dominant intensity modes, 2) multi-peak detection for handling complex outdoor backgrounds with sky, clouds, and terrain, 3) explicit overexposure attenuation to mitigate bright sky effects, and 4) computational efficiency through histogram-based processing independent of image resolution. We evaluate the proposed method against 12 existing thresholding methods on 116 benchmark images, demonstrating superior performance with precision of 81.4% (11.66% higher than the nearest competitor), recall of 94.0% (11.37% improvement), and coverage of 61.4% (48.31% improvement). The method achieves processing times of 0.817 ms at 4K resolution (3840×2160), delivering 27.4× speedup over the nearest quality competitor while maintaining exceptional detection accuracy, making it ideal for real-time drone surveillance in resource-constrained embedded systems.

INDEX TERMS Drone detection, Fourier transform, histogram approximation, adaptive thresholding, image segmentation, real-time processing.

I. INTRODUCTION

Uncrewed Aerial Vehicles (UAVs), or drones, have revolutionized various sectors, from aerial surveys and logistics to environmental monitoring. However, their proliferation also introduces significant risks, including invasions of privacy, violations of airspace, and potential threats to critical infrastructure. Consequently, robust and efficient drone detection systems are vital for improving security in both civilian and military contexts. Among detection modalities, vision-based approaches stand out for their cost-effectiveness and integration potential with existing camera networks; yet, they must contend with challenges such as real-time processing demands, environmental variability, and the need for deployment on resource-constrained platforms.

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Current drone detection techniques span multiple modalities including radar, radio-frequency (RF) sensing, acoustic analysis, and optical methods. Vision-based strategies, in particular, leverage image-processing pipelines that range from classical edge detection and feature extraction to modern machine-learning models such as YOLO or Convolutional Neural Networks (CNNs). While deep learning offers high accuracy, it incurs substantial computational overhead and latency, often requiring GPU acceleration and exceeding processing budgets for high-frame-rate systems operating at 200 fps or more. Furthermore, deep learning approaches demand extensive labeled training data and struggle with generalization to novel environments without retraining.

Simpler contrast-based methods provide a lightweight alternative by converting images to grayscale and applying thresholding to isolate low-intensity objects such as drones

against brighter backgrounds (sky, clouds). These approaches are particularly attractive for embedded deployment on edge devices and scenarios requiring sub-millisecond processing without specialized hardware. However, the efficacy of contrast-based methods hinges critically on threshold selection, and many traditional algorithms—including Otsu, Triangle, Minimum, and Yen—struggle with the noisy, multi-modal histograms typical of outdoor scenes featuring varying lighting conditions, overexposed sky regions, and heterogeneous backgrounds combining clouds, terrain, and structures.

To address these limitations, this paper introduces a novel Fourier-based adaptive thresholding technique specifically designed for outdoor drone detection. The core innovation lies in applying the Fast Fourier Transform (FFT) to the image histogram and retaining only low-frequency components (typically 6–10 harmonics). This approach transforms the discrete, noise-corrupted histogram into a smooth continuous function that preserves dominant background intensity modes while suppressing high-frequency noise from texture variations, ISO noise, and compression artifacts. The smoothed histogram enables precise multi-peak detection, allowing the algorithm to identify and exclude multiple background components (sky, clouds, ground) rather than assuming a simple bimodal distribution as in classical methods.

The proposed method introduces several additional innovations beyond Fourier smoothing. An overexposure attenuation parameter (`ovexp_k`) explicitly handles bright sky regions by suppressing histogram bin 255 before transformation, delivering a 62.08% false positive reduction on overexposure-heavy imagery while maintaining robust performance on well-exposed scenes. Multi-modal background detection through iterative peak search (`nlp`, `npr`, `pli` parameters) captures complex outdoor backgrounds that defeat single-threshold methods. A precision-recall tuning mechanism (`lad` parameter) enables application-specific optimization of the detection operating point without algorithmic modification, providing 51.58% false positive reduction when configured for high-security scenarios. Finally, architectural decoupling—performing downsampling before histogram analysis and applying FFT to the fixed-size 256-bin histogram—achieves processing time largely independent of input resolution, enabling sub-millisecond operation even at 4K resolution.

In contrast to classical histogram-based thresholding, the proposed method combines three elements that, to the best of our knowledge, have not previously been integrated in a single pipeline for small-drone detection: (i) aggressive low-pass Fourier approximation of the 256-bin histogram to obtain a smooth, resolution-independent representation of the background intensity distribution, (ii) explicit modeling of multi-modal outdoor backgrounds through iterative peak search with tunable parameters, and (iii) dedicated overexposure attenuation and precision-recall control designed around drone-in-sky surveillance scenarios. The ablation study in Section IV empirically demonstrates that each of these

components contributes significantly to the final performance and latency profile.

Comprehensive evaluation on 116 benchmark images across 13 thresholding methods demonstrates substantial quantitative advantages. Compared to the nearest quality competitor (Minimum method), the proposed approach achieves 11.66% higher precision (0.814 vs 0.729), 11.37% higher recall (0.940 vs 0.844), and 48.31% higher coverage (0.614 vs 0.414), indicating superior detection accuracy with tighter bounding boxes and fewer false alarms. Speed benchmarks at 4K resolution (3840×2160) reveal $27.4\times$ speedup over Minimum method, $26.4\times$ over Yen, and $30.3\times$ over Intermodes, with processing time of 0.817 ms per frame enabling 1,224 fps throughput on the test platform (Apple M3 Pro). These combined quality and speed improvements position the proposed method as a practical solution for real-world outdoor surveillance deployments requiring real-time processing on embedded or edge computing platforms.

The remainder of this paper is organized as follows. Section II reviews related work on thresholding techniques and drone detection approaches. Section III details the proposed method, including the intuition behind the approach, preprocessing steps, Fourier-based histogram approximation, multi-modal threshold selection, and contour clustering. Section IV presents comprehensive ablation studies analyzing the contribution of each parameter to overall performance. Section V presents experimental results, comparing performance and quality metrics against 12 existing thresholding methods and discussing limitations and applicability to noisy and thermal imaging scenarios. Finally, Section VI concludes with insights on the method's novelty, quantitative advantages, practical implications, and directions for future research.

II. RELATED WORK

Thresholding is a fundamental technique in image segmentation, with numerous methods proposed over several decades. Otsu [1] introduced a seminal automatic method by maximizing the between-class variance in the histogram, assuming a bimodal distribution. While widely used, Otsu's method struggles with multi-modal histograms common in outdoor scenes and is sensitive to noise.

The intermodes and minimum threshold methods assume a bimodal histogram and apply smoothing until two peaks remain, selecting the valley between them [2]. These methods work well for simple foreground-background scenarios but fail when multiple background modes exist (e.g., sky and clouds) or when noise obscures the valley.

The IsoData algorithm uses an iterative approach to find the threshold as the average of the mean intensities of two classes [3]. Li's method minimizes the cross-entropy between the original image and the thresholded version [4]. The maximum entropy method selects the threshold that maximizes the sum of entropies of foreground and background distributions [5]. While these information-theoretic

approaches demonstrate robustness to certain histogram characteristics, they remain susceptible to noise and lack explicit multi-modal handling.

Simpler methods include the mean method, which uses the average intensity as threshold [6], and the percentile method, which sets the threshold based on a fixed pixel percentage. The minimum error method assumes Gaussian mixture distributions and minimizes classification error [7]. The moments method preserves histogram moments in the thresholded image [8]. The triangle method constructs a line from the histogram peak to the end and selects the maximum distance point [9]. The Yen method employs a correlation and entropy-based criterion [10]. None of these methods explicitly address the multi-modal backgrounds and overexposure challenges prevalent in outdoor drone detection.

Fourier transforms have been extensively applied in image processing for frequency-domain filtering and enhancement [11]. Application of Fourier analysis to histograms for spectral smoothing has been explored in statistical distribution analysis [12], but its use for adaptive thresholding in outdoor object detection, particularly with multi-peak detection and overexposure handling, represents a novel contribution. The proposed method uniquely combines Fourier-based histogram smoothing with multi-modal background peak detection, a combination absent from existing literature.

Recent advances in drone detection predominantly employ deep learning models such as YOLO [13] and variants of CNNs, achieving high accuracy at the cost of substantial computational requirements (GPU acceleration, high memory footprint, milliseconds to seconds of latency per frame). These approaches excel in complex scenarios but are impractical for embedded deployment or applications requiring sub-millisecond latency. Our method offers a complementary lightweight alternative suitable for resource-constrained environments, achieving competitive detection quality through intelligent signal processing rather than learned features.

Beyond heavy general-purpose detectors, several works explore lightweight deep and hybrid pipelines for small-object surveillance, combining motion-based region proposals or background subtraction with compact CNN or YOLO variants to reduce the effective inference cost. These systems typically operate in a latency regime of several milliseconds to tens of milliseconds per frame [22] on GPU-accelerated platforms, achieving strong accuracy but still relying on trained models, significant memory footprint, and hardware acceleration. In contrast, the method proposed in this paper is a purely analytical, training-free thresholding approach designed to run at sub-millisecond latency on a single CPU core, making it suitable for extreme real-time applications and ultra-low-power embedded deployments where even lightweight deep models remain too expensive.

Contour clustering post-thresholding, as employed in our pipeline, draws from connected component analysis [14]. The clustering step addresses a practical challenge in outdoor

detection: partial occlusion, blur, and overexposure can fragment a single drone into multiple detected regions. Our spatial proximity-based clustering merges these fragments into cohesive detections, improving recall and reducing false negatives from partial observations.

III. PROPOSED METHOD

The proposed algorithm operates on grayscale images for drone detection via contrast. The core innovation lies in using Fourier transform to approximate the histogram and derive an adaptive threshold. As illustrated in Figure 1, typical input



FIGURE 1. Example input image from the dataset showing a drone in an outdoor scene.

images feature drones as small, low-intensity objects against brighter backgrounds like skies. In the center of the image there are four small birds, one drone and white clouds on the top.

A. IMAGE PREPROCESSING

The input image frame is first converted to grayscale. To optimize computation and ensure efficient performance, especially for high-resolution images, the image is scaled down if its longest side exceeds a threshold `scale_px` pixels:

$$\text{scale_factor} = \max \left(1, \text{round} \left(\frac{\max(w, h)}{\text{scale_px}} \right) \right), \quad (1)$$

where w and h are the width and height of the frame. The scaled region of interest (ROI) is then:

$$\text{scaled_roi} = \text{frame}[:, \text{scale_factor} :: \text{scale_factor}]. \quad (2)$$

This downsampling is necessary to reduce computational overhead, as histogram computation on full-resolution images can be time-intensive for large frames. Importantly, it does not affect the quality of threshold selection, since the histogram is primarily used to detect the overall image background and dominant grayscale distributions, rather than fine details such as small objects like drones. This scaled ROI is then used for histogram computation, as detailed next.

B. HISTOGRAM COMPUTATION AND FOURIER APPROXIMATION

The histogram $h(i)$ for $i = 0, 1, \dots, 255$ is computed as the count of pixels with intensity i :

$$h(i) = \sum_{p \in \text{scaled_roi}} \delta(p, i), \quad (3)$$

where δ is the Kronecker delta. To mitigate overexposure effects, the highest bin is attenuated by a factor ovexp_k :

$$h(255) \leftarrow h(255) \times \text{ovexp_k}. \quad (4)$$

The real-valued Fast Fourier Transform (RFFT) is applied to the histogram:

$$f(k) = \text{RFFT}(h), \quad k = 0, 1, \dots, 128, \quad (5)$$

retaining only the low-frequency components by zeroing out higher harmonics (beyond the first harmonics_n):

$$f(k) = 0 \quad \forall k > \text{harmonics_n}. \quad (6)$$

The approximated histogram is reconstructed via the inverse RFFT:

$$\hat{h}(i) = \text{IRFFT}(f), \quad i = 0, 1, \dots, 255. \quad (7)$$

This low-pass filtering smooths the histogram, emphasizing dominant modes while suppressing noise. As shown in Figure 2, the approximation smooths out minor fluctuations while preserving the main peaks.

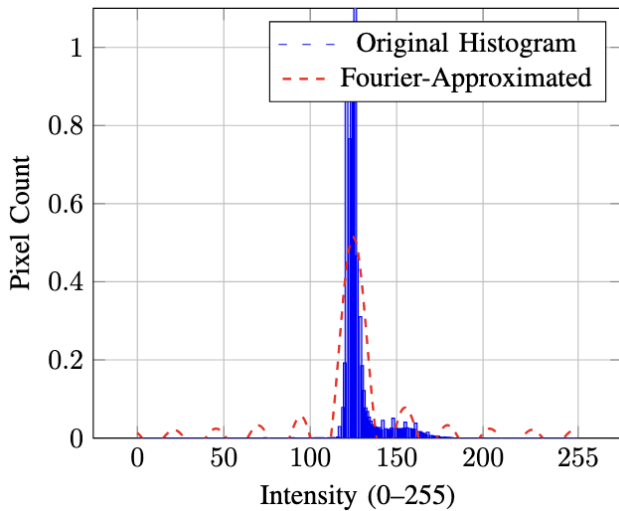


FIGURE 2. Original vs. Fourier-approximated histogram for a sample image. The Fourier method uses only the first 10 low-frequency components to smooth the histogram, preserving dominant background while suppressing noise.

C. THRESHOLD SELECTION

The threshold is derived by identifying the dominant grayscale range in $\hat{h}$. First, locate the primary peak:

$$\text{main_idx} = \text{argmax}_i \hat{h}(i). \quad (8)$$

Initialize borders at this index: $\text{left} = \text{main_idx}$, and $\text{pmax} = \hat{h}(\text{main_idx})$. Extend the left border backward until

the intensity drops below the peak amplitude multiplied by pzt :

$$\text{while } \text{left} > 0 \text{ and } \hat{h}(\text{left}) > \text{pmax} \times \text{pzt} : \text{left} -- = 1. \quad (9)$$

Refine by searching for up to nlp additional peaks to the left (beyond index pli to focus on darker regions):

For each of nlp iterations, if $\text{left} > \text{pli}$:

- Scan leftward to find a new local maximum.
- If the new max exceeds npr of pmax , update left and pmax .
- Further extend the left border until the intensity falls below pzt of the current max pmax

Finally the low threshold is adjusted by the lad parameter: $\text{low} = \text{left} + \text{lad}$. The binary segmentation is applied:

if $0 \leq \text{frame}(p) \leq \text{low}$ then

$$\text{binary}(p) = 1 \quad \text{otherwise} \quad \text{binary}(p) = 0 \quad (10)$$

This marks by 1 the darker regions pixels (e.g., drones against bright skies) for further contour detection processing.

D. CONTOUR DETECTION AND CLUSTERING

Contours are extracted from the binary image using external retrieval and simple chain approximation. Each contour is bounded by a rectangle, and parameters (size in cm, angular position) are computed using field-of-view (FOV) and distance conversions:

$$w_{\text{deg}} = \text{pts_to_deg} \left(\frac{w}{\text{width}}, \text{FOV}_x \right), \quad (11)$$

$$\text{size_cm} = \max(\text{deg_to_cm}(w_{\text{deg}}, d), \text{deg_to_cm}(h_{\text{deg}}, d)) \quad (12)$$

filtering by $\text{min_size} \leq \text{size_cm} \leq \text{max_size}$. Nearby contours are clustered using a distance matrix (Euclidean on centers) and a pixel threshold derived from max object size. Clusters are merged into encapsulating boxes, forming UAV objects.

E. METHOD INTUITION

The fundamental insight driving this approach stems from the characteristic appearance of drones in outdoor surveillance scenarios. A drone flying in the sky typically appears as a dark, low-contrast object against a predominantly homogeneous background that occupies most of the image frame. This background exhibits dominant color tones—often bright sky, clouds, or a combination thereof—and may contain substantial overexposed regions (intensity 255) from direct sunlight.

The histogram of such an image reveals the distribution of pixel intensities, manifesting as a graph with distinct peaks. The largest peaks correspond to the background colors that dominate the frame, which we must identify and remove to isolate the drone. However, in overexposed scenes, the maximum peak often occurs at intensity 255 (pure white), masking the true background distribution. To address this, we initially attenuate bin 255 by multiplying it by a factor,

enabling detection of the actual background modes beneath the overexposure noise.

Critically, the maximum peak does not always represent the entire background. Consider a scene with sky and clouds: the sky may occupy the majority of pixels (producing the tallest peak), while clouds form a secondary, shorter peak. Both constitute background that must be excluded. The algorithm therefore searches for the primary peak to identify the brighter-than-drone background components, then iteratively locates secondary peaks to the left to capture additional background modes such as mid-tone clouds or terrain.

Once all relevant background peaks are identified, the threshold is determined by finding the lowest leftmost edge of the leftmost peak. This boundary represents the intensity level below which pixels transition from background to foreground. Applying this threshold produces a binary mask: background regions (above the threshold) are marked white, while darker objects including drones (below the threshold) are marked black.

The Fourier transform serves two critical functions in this process. First, it converts the discrete histogram values into a continuous mathematical function, enabling smooth peak detection and boundary identification. Second, by retaining only low-frequency components, it filters out high-frequency noise from the histogram while preserving the dominant background structure. This low-pass filtering both improves robustness to histogram irregularities and reduces computational cost by limiting the number of frequency components processed.

Following thresholding, contour detection identifies candidate object regions. Since drones may be partially occluded, blurred, or fragmented by overexposure, the algorithm employs contour clustering to merge nearby detected fragments. This spatial grouping “assembles” disconnected parts of a single drone into one cohesive detection rectangle, improving detection completeness and reducing false negatives from partial observations.

IV. ABLATION STUDY

To validate our parameter choices and design decisions, we conducted a comprehensive ablation study evaluating the contribution of each component to the algorithm’s performance. The study examined eight hyperparameters: `harmonics_n`, `lad`, `nlp`, `npr`, `ovexp_k`, `pli`, `pzr`, and `scale_px`, as well as the fundamental choice between FFT-based interpolation and raw discrete histogram data.

A notable observation from the ablation study is that performance varies smoothly around the chosen parameter configuration: for most parameters, a relatively broad range of values yields similar TPPI, FPPI, and coverage, rather than exhibiting a narrow, brittle optimum. This indicates that the method does not rely on fragile fine-tuning and that the selected settings should transfer reasonably well to unseen but related datasets. In practice, a small calibration set could be used to automatically choose or refine parameters via grid

search or simple optimization, further improving robustness without manual tuning.

A. EXPERIMENTAL SETUP

Following standard ablation methodology, we varied only one parameter per experiment while keeping all others fixed at their baseline values. This approach enables isolated assessment of each parameter’s impact on algorithm performance. The evaluation dataset consists of the 116 benchmark images used for results evaluation in Section V, containing an average of 1.034483 total objects per image (TOPI).

B. EVALUATION METRICS

We assessed performance using four metrics:

- **TPPI** (True Positives Per Image): measures detection sensitivity, equivalent to the number of correctly detected drones per image
- **FPPI** (False Positives Per Image): quantifies the false alarm rate, representing spurious detections per image
- **Coverage**: represents the average Intersection over Union (IoU) between the detected bounding rectangles and the ground-truth bounding boxes for the true positive detections, expressed as a ratio
- **Processing Time**: computational efficiency measured in milliseconds

C. FFT-BASED INTERPOLATION VS. RAW HISTOGRAM DATA

A fundamental design choice in our method is the use of FFT-based interpolation to smooth the histogram before threshold selection. To validate this decision, we compared the proposed FFT-based approach against a baseline using only raw discrete histogram values without any interpolation or smoothing.

Table 1 presents the performance comparison between these two approaches.

TABLE 1. Performance comparison: FFT-based interpolation vs. raw histogram data.

Method	TPPI	FPPI	Coverage	Time (ms)
With FFT	0.888	0.871	0.594	0.112
Without FFT	0.888	9.276	0.510	0.245

The FFT-based approach demonstrates substantial advantages over raw histogram data. While maintaining identical TPPI (0.888), the FFT method reduces FPPI by **813.0%** relative to TOPI (from 9.276 to 0.871, representing a 90.6% reduction in absolute terms). This dramatic reduction in false positives is critical for real-world surveillance applications where false alarms erode system trust and operational efficiency. Additionally, FFT achieves a **54.1%** reduction in processing time (from 0.245 ms to 0.112 ms) and improves coverage by **16.5%** (from 0.510 to 0.594).

These results confirm that FFT-based interpolation is essential for both accuracy and computational efficiency.

The smoothing effect of low-frequency Fourier components effectively filters histogram noise from minor pixel variations and lighting irregularities, enabling more robust identification of the dominant background intensity ranges. Without this smoothing, the threshold selection algorithm becomes sensitive to spurious local peaks in the raw histogram, leading to the observed explosion in false positives.

D. NUMBER OF FOURIER HARMONICS

The `harmonics_n` parameter controls the number of low-frequency harmonic components retained in the FFT-based histogram approximation. As described in Section III-B, we apply a low-pass filter by zeroing out frequency components beyond the first `harmonics_n` harmonics. This parameter represents a fundamental trade-off: too few harmonics may over-smooth the histogram and lose important distributional information, while too many may retain high-frequency noise that degrades threshold selection.

We systematically evaluated `harmonics_n` values ranging from 1 to 128 to determine the optimal setting. Figure 3 illustrates the performance trends across this parameter range, showing that detection quality peaks at low values and gradually degrades as the number of harmonics increases.

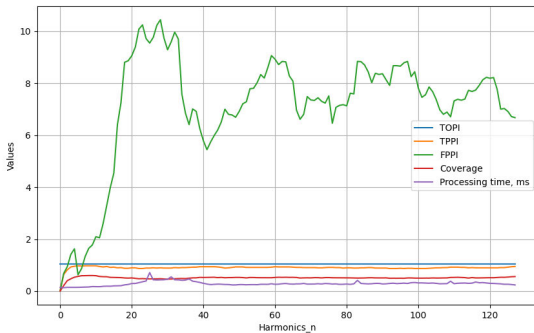


FIGURE 3. Performance metrics across different `harmonics_n` values. The plot shows TPPI, FPPI, TOPI (reference baseline), Coverage, and Processing Time for values ranging from 0 to 128. Performance peaks at `harmonics_n` = 6.

Table 2 compares the optimal `harmonics_n` value against the maximum value tested (`harmonics_n` = 128), which effectively represents the case where harmonic limiting is not applied.

TABLE 2. Performance comparison: optimal vs. no-limiting `harmonics_n` values.

Config.	TPPI	FPPI	Coverage	Time (ms)
$n = 6$ (optimal)	0.966	0.871	0.594	0.113
$n = 128$ (no limit)	0.922	6.914	0.541	0.212

The optimal value of `harmonics_n` = 6 significantly outperforms the no-limiting configuration. Compared to `harmonics_n` = 128, the optimal setting achieves a **4.67%**

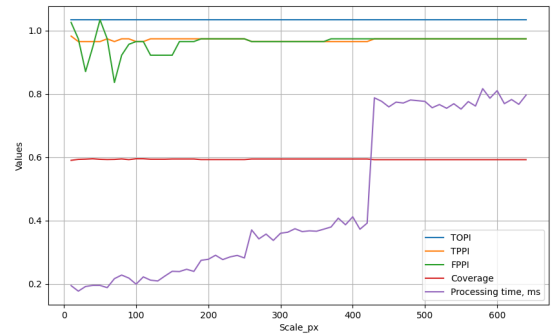


FIGURE 4. Performance metrics across different `scale_px` values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from 10 to 640 pixels.

improvement in TPPI (from 0.922 to 0.966) and a dramatic **87.41%** reduction in FPPI (from 6.914 to 0.871). Coverage improves by **9.70%** (from 0.541 to 0.594), demonstrating substantially better localization of detected objects. The processing time decreases by **46.86%** (from 0.212 ms to 0.113 ms), providing both accuracy and efficiency gains.

These results reveal a critical insight: excessive harmonic components introduce noise and spurious detections rather than improving signal representation. The low optimal value suggests that the dominant frequency characteristics of outdoor scene histograms are captured by a very small number of low-frequency harmonics corresponding to broad intensity distributions (sky, ground, shadows). Higher-order components like small objects, birds, drones, ISO noise or image compression artifacts primarily encode high-frequency noise from minor pixel variations that is detrimental to robust background threshold selection rather than beneficial.

For our drone detection application, where the goal is to identify the dominant background intensity range, the first 6 harmonics suffice to capture the essential bimodal or multimodal structure of typical outdoor histograms while filtering out the fine-scale irregularities that lead to false positives. This validates our design choice and demonstrates that aggressive low-pass filtering is not only acceptable but optimal for histogram-based thresholding in cluttered outdoor scenes.

E. IMAGE DOWNSAMPLING FOR HISTOGRAM COMPUTATION

The `scale_px` parameter controls the maximum dimension (width or height) to which input images are downsampled before histogram computation, as described in Section III-A. This downsampling is performed exclusively for the histogram calculation step and does not affect the resolution at which contour detection operates. The parameter represents a fundamental trade-off: aggressive downsampling reduces computational cost but may degrade histogram fidelity, while processing at full resolution provides the most accurate histogram at the expense of processing time.

TABLE 3. Performance comparison: optimal downsampling vs. no downsampling.

Config.	TPPI	FPPI	Coverage	Time (ms)
$s = 30$ (optimal)	0.966	0.871	0.594	0.192
$s = 640$ (no downsamp.)	0.974	0.974	0.592	0.796

We systematically evaluated `scale_px` values ranging from 10 to 640 pixels to determine the contribution of the downsampling to overall method performance. Figure 4 illustrates how performance metrics vary across this range, revealing that detection quality remains relatively stable for moderate downsampling but processing time increases dramatically without downsampling.

Table 3 compares the optimal downsampling configuration against no downsampling.

The results demonstrate the significant contribution of downsampling to both computational efficiency and detection accuracy. Comparing optimal downsampling (`scale_px` = 30) against no downsampling (`scale_px` = 640), we observe a dramatic **75.94%** reduction in processing time (from 0.796 ms to 0.192 ms). Remarkably, this massive speed improvement comes with minimal quality cost: TPPI decreases by only 0.88% (from 0.974 to 0.966), while FPPI actually improves by **10.62%** (from 0.974 to 0.871). Coverage remains essentially identical at 0.594.

This counterintuitive result—where downsampling improves FPPI while maintaining TPPI—reveals an important property of histogram-based threshold selection. By reducing image resolution before histogram computation, the downsampling operation effectively performs spatial averaging that suppresses high-frequency texture noise and minor pixel variations. This smoothing helps the histogram better represent the dominant intensity distributions (sky, ground, clouds) while filtering out spurious local intensity variations and small objects that can mislead background threshold selection in full-resolution histograms.

The stability of detection quality across downsampling levels validates our hypothesis from Section III-A that histogram-based threshold selection depends primarily on global intensity distributions rather than fine spatial details. The histogram's statistical properties—the relative proportions of pixels in different intensity ranges—remain largely invariant under moderate downsampling because the overall scene composition (ratios of sky, ground or clouds) does not fundamentally change when reducing resolution. This robustness makes aggressive downsampling not only acceptable but beneficial for this application.

The **75.94%** processing time reduction enabled by downsampling is critical for real-time performance, particularly when processing high-resolution video streams. For a 1920×1080 input image, computing the histogram at full resolution requires processing over 2 million pixels, whereas downsampling to 30 pixels (maintaining aspect ratio) processes only approximately 900 pixels—a reduction of over 2000 $\times$. This computational savings translates

directly to the observed $4\times$ speedup in processing time, enabling sub-millisecond histogram computation even on resource-constrained embedded platforms.

These findings confirm that aggressive downsampling for histogram computation is not only acceptable but optimal for drone detection. The chosen approach of computing histograms on heavily downsampled images (30–400 pixels) while performing contour detection on full-resolution thresholded images provides the best of both worlds: fast, robust threshold selection from global intensity distributions, combined with precise object localization at full resolution. This design choice is particularly valuable for embedded and edge computing scenarios where computational resources are constrained and real-time performance is essential.

F. PEAK LEFT INDEX FOR SECONDARY PEAK SEARCH

The `pli` (Peak Left Index) parameter determines the leftmost histogram index from which the algorithm begins searching for secondary peaks, as described in Section III-C. This parameter controls the focus region for identifying darker intensity ranges that may correspond to drone objects. By restricting the search to indices less than `pli`, the algorithm avoids incorporating bright background regions into the threshold calculation, effectively focusing on low-intensity foreground objects against brighter backgrounds.

We systematically evaluated `pli` values ranging from 0 to 254 to determine the optimal setting. Figure 5 illustrates how performance metrics vary across this parameter range.

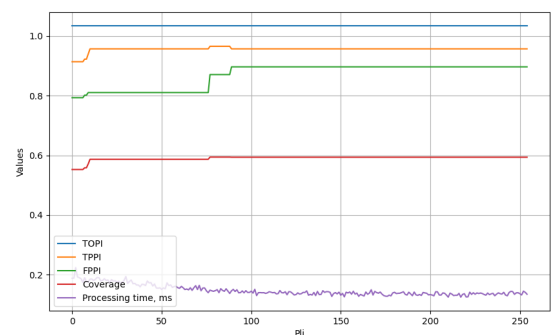
**FIGURE 5.** Performance metrics across different `pli` values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from 0 to 254. Peak performance occurs around `pli` = 88.

Table 4 compares the optimal `pli` value against the no-limiting configuration (`pli` = 254, where virtually all histogram bins are searchable).

TABLE 4. Performance comparison: optimal vs. no-limiting `pli` values.

Config.	TPPI	FPPI	Coverage	Time (ms)
$p = 88$ (optimal)	0.966	0.871	0.594	0.137
$p = 254$ (no limit)	0.957	0.897	0.594	0.134

The optimal value of `pli` = 88 demonstrates measurable improvements over the no-limiting configuration. Compared

to $pli = 254$, the optimal setting achieves a **0.90%** improvement in TPPI (from 0.957 to 0.966) and a **2.88%** reduction in FPPI (from 0.897 to 0.871). Coverage remains essentially identical (0.594). The processing time increases marginally by 1.97% (from 0.134 ms to 0.137 ms), representing a negligible trade-off.

These results reveal that restricting the secondary peak search to exclude the darkest histogram regions (below index 88) provides a tangible benefit for drone detection accuracy. The fundamental purpose of this parameter is to ensure that the secondary peak search identifies only **background intensity modes** rather than the **drone objects themselves**. Since drones typically appear as darker objects than the background in outdoor scenes, setting a lower boundary at $pli = 88$ prevents the algorithm from mistakenly selecting a peak that represents the drone intensity distribution. This ensures the threshold selection clearly separates background from foreground, rather than attempting to threshold within or around the drone's intensity range.

The mechanism works as follows: by excluding histogram regions below index 88 from the secondary peak search, the algorithm is forced to find secondary peaks only in the mid-to-bright intensity ranges that correspond to various background elements (ground, clouds, structures) rather than the darker drone objects. This produces a threshold that cleanly cuts the background away from the scene while preserving the darker foreground drones. If pli were set too low (e.g., 0–50), the algorithm could potentially identify the drone's intensity peak as a “secondary peak,” leading to a threshold that incorrectly classifies parts of the drone as background or creates ambiguous boundary regions.

The optimal value of 88 corresponds to approximately one-third of the 8-bit intensity range, suggesting that for typical outdoor drone detection scenarios, drones occupy intensity ranges below this threshold while background elements occupy ranges above it. This aligns with practical observations: in outdoor surveillance with bright skies, drones appear significantly darker than most background elements, typically falling in the 0–87 intensity range due to their silhouette appearance against bright backgrounds. The secondary peak search in the 88–255 range then identifies background intensity modes (sky, ground, clouds), enabling clean foreground-background separation.

Values significantly below 88 (e.g., $pli = 0$ –50) degrade performance because they risk including the drone's intensity distribution in the secondary peak search, leading to thresholds that cut into the object itself rather than around it. Values significantly above 88 (e.g., $pli > 150$) also show diminishing returns because they overly restrict the search space, potentially missing relevant background peaks in moderately bright regions and forcing the algorithm to rely solely on the primary (brightest) peak.

Figure 6 illustrates a practical example of secondary peak detection in a challenging scenario with a bitone background. The scene contains both dark trees and a yellowish field,

creating two distinct background intensity modes visible as two prominent peaks in the histogram. The algorithm successfully identifies both background peaks (marked in the histogram visualization), enabling threshold selection that separates the even darker drone (shown in the yellow bounding box) from both background components. While this example does not imply that the method performs optimally on all such complicated backgrounds, it demonstrates the capability of the secondary peak mechanism to handle multi-modal background distributions when appropriate parameters are set.

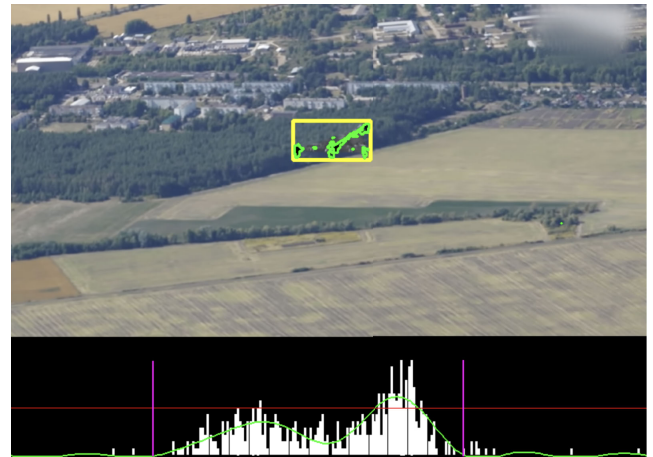


FIGURE 6. Example of secondary peak detection with bitone background. **Top:** Detection result showing a dark drone (yellow box) against a background with two distinct intensity modes (dark trees and yellowish field). **Bottom:** Histogram visualization showing two background peaks (main peak for bright field, secondary peak for darker trees) that enable successful foreground-background separation with the drone appearing darker than both background components.

G. NUMBER OF SECONDARY PEAK SEARCH ITERATIONS

The nlp (Number of Left Peaks) parameter controls how many iterations the algorithm performs when searching for secondary peaks to the left of the main histogram peak, as described in Section III-C. Each iteration attempts to find an additional local maximum in the darker intensity regions that corresponds to different background intensity modes, enabling the algorithm to handle multi-modal backgrounds (such as the bitone example in Figure 6 with both dark trees and bright field). The parameter represents a trade-off between the ability to capture complex multi-tone backgrounds and computational cost.

We systematically evaluated nlp values ranging from 0 to 9 to determine the optimal setting. Figure 7 illustrates how performance metrics vary across this parameter range.

Table 5 compares configurations with and without secondary peak search enabled.

The results demonstrate clear benefits of enabling secondary peak search. Compared to $nlp = 0$, any configuration with $nlp \geq 1$ achieves a **0.90%** improvement in TPPI (from 0.957 to 0.966) and a **2.88%** reduction in FPPI (from 0.897 to 0.871). Coverage remains essentially identical (0.594), while

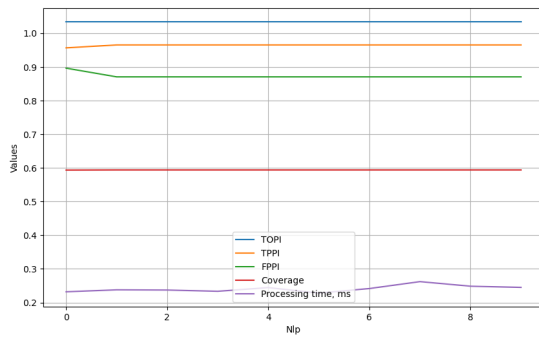


FIGURE 7. Performance metrics across different nlp values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from 0 to 9.

TABLE 5. Performance comparison: with vs. without secondary peak search.

Config.	TPPI	FPPI	Coverage	Time (ms)
$n \geq 1$ (with search)	0.966	0.871	0.594	0.228–0.262
$n = 0$ (no search)	0.957	0.897	0.594	0.232

processing time varies minimally between 0.228 ms and 0.262 ms across different nlp values.

The jump in performance from $nlp = 0$ to $nlp \geq 1$ is significant. Without secondary peak search ($nlp = 0$), the algorithm relies solely on the primary histogram peak to define the threshold boundaries. This works reasonably well when the histogram exhibits a simple unimodal background distribution. However, in complex outdoor scenes with multi-modal backgrounds (multiple background intensity modes such as sky, ground, trees, shadows, buildings), restricting to only the primary peak can lead to suboptimal threshold selection. The algorithm may set a threshold based only on the brightest background component (e.g., sky), potentially misclassifying darker background elements (e.g., trees, ground) as foreground, which increases false positives.

Enabling even a single secondary peak search iteration ($nlp \geq 1$) allows the algorithm to identify and incorporate additional background intensity modes into the threshold calculation. For instance, in a scene with both bright sky and darker ground terrain, the secondary peak search can identify the ground's intensity distribution, enabling a threshold that properly excludes both background components. This refinement reduces false positives by ensuring that all major background modes are considered, not just the brightest one, as evidenced by the consistent FPPI reduction from 0.897 to 0.871.

Given that all values from $nlp = 1$ to 9 produce identical detection quality, the choice of specific value depends on robustness considerations rather than performance optimization. A moderate value such as $nlp = 2$ –5 provides a safety margin for unusual histogram configurations that might require multiple iterations to converge, while avoiding unnecessarily high iteration counts. The processing time

variation across $nlp = 1$ –9 is minimal (14.9% spread), making computational cost essentially irrelevant to this choice.

H. NEXT PEAK RATIO THRESHOLD

The npr (Next Peak Ratio) parameter determines the minimum relative height that a secondary peak must reach to be considered significant during the threshold refinement process, as described in Section III-C. Specifically, a newly discovered peak is only incorporated into the threshold calculation if its amplitude exceeds npr times the current peak maximum. This parameter controls the sensitivity of the algorithm to minor peaks versus major intensity modes in the histogram.

We systematically evaluated npr values ranging from 0.0 to 0.99 to determine the optimal setting. Figure 8 illustrates how performance metrics vary across this parameter range, revealing a fundamental trade-off between detection sensitivity and false alarm rate.

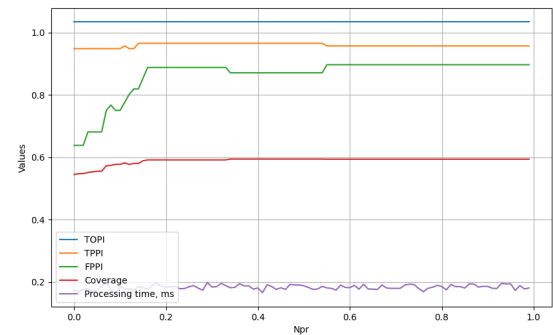


FIGURE 8. Performance metrics across different npr values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from 0.0 to 0.99.

Table 6 compares the optimal npr configuration against the baseline case where all secondary peaks are accepted regardless of height ($npr = 0$).

TABLE 6. Performance comparison: optimal vs. no peak height filtering.

Config.	TPPI	FPPI	Coverage	Time (ms)
$r = 0.34$ (optimal)	0.966	0.871	0.594	0.181
$r = 0.00$ (no threshold)	0.948	0.638	0.545	0.174

The results demonstrate the contribution of the npr parameter in filtering secondary background peaks. Comparing the baseline ($npr = 0$, accepting all peaks regardless of height) against the optimal configuration ($npr = 0.34$), we observe substantial improvements: TPPI increases by **1.90%** (from 0.948 to 0.966), coverage improves by **9.09%** (from 0.545 to 0.594), though FPPI also increases by **36.53%** (from 0.638 to 0.871). This trade-off reveals the parameter's fundamental role: without any threshold ($npr = 0$), the algorithm accepts all secondary peaks indiscriminately, including numerous spurious peaks from histogram noise. Paradoxically, this overly permissive acceptance results in lower false positives

but also poorer detection performance and localization quality.

The mechanism underlying this counterintuitive behavior becomes clear when considering what happens with unrestricted peak acceptance. At $npr = 0$, the algorithm accepts any peak in the darker histogram regions as a potential background mode, including very small peaks caused by noise, texture variations, or isolated pixel groups. These spurious minor peaks cause the threshold boundaries to contract inappropriately, creating overly restrictive threshold regions that exclude areas containing actual objects. The result is conservative threshold selection despite accepting all peaks—the algorithm effectively “over-fits” to histogram noise rather than capturing genuine background structure. By requiring peaks to exceed a minimum relative height ($npr = 0.34$), the algorithm filters out these spurious minor peaks, focusing only on genuine multi-modal background distributions like the bitone example in Figure 6. This selective approach prevents threshold over-contraction, allowing the algorithm to set broader, more appropriate threshold boundaries that capture objects more effectively.

At the optimal threshold ($npr = 0.34$), the algorithm achieves the best balance between detection performance and false alarm control. TPPI reaches 0.966 and coverage improves to 0.594, indicating better detection sensitivity and localization accuracy. The increase in FPPI to 0.871, while notable compared to the baseline’s 0.638, represents an acceptable trade-off: the algorithm is now detecting more objects overall (+1.90% TPPI) with substantially better bounding box quality (+9.09% coverage), at the expense of some additional false alarms. This configuration effectively filters histogram noise while still capturing genuine multi-modal backgrounds.

The choice of npr depends on the application’s priorities. For surveillance applications prioritizing minimal false alarms (e.g., high-security scenarios), the baseline $npr = 0$ may be preferable, accepting lower detection rate (TPPI = 0.948) and coverage (0.545) in exchange for fewer false positives (FPPI = 0.638). For applications prioritizing detection completeness and localization quality, the optimal value ($npr = 0.34$) maximizes performance despite higher false alarm rates. The data indicates that peak height filtering provides substantial contribution to detection quality, justifying the increased FPPI through significant gains in detection sensitivity and bounding box accuracy.

Processing time remains relatively stable (0.174–0.181 ms), indicating that this parameter primarily affects peak selection quality rather than computational cost.

I. PEAK ZERO RATIO

The p_zr (Peak Zero Ratio) parameter controls when the algorithm stops extending the left border during threshold selection, as described in Section III-C. Specifically, the leftward extension terminates when the histogram intensity drops below p_zr times the peak maximum. This parameter determines how aggressively the threshold expands into

darker regions: lower values allow more expansion (capturing more potential foreground objects but risking false positives), while higher values restrict expansion (reducing false alarms but potentially missing detections).

We systematically evaluated p_zr values ranging from -0.20 to 0.19 to determine the contribution of this parameter to overall method performance. Figure 9 illustrates how performance metrics vary across this parameter range.

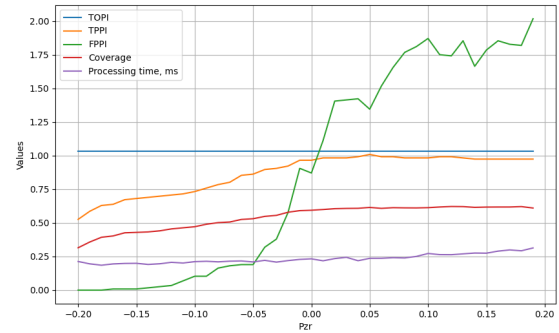


FIGURE 9. Performance metrics across different p_zr values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from -0.20 to 0.19 .

Table 7 compares the optimal p_zr configuration against the baseline case ($p_zr = 0$).

TABLE 7. Performance comparison: optimal vs. baseline border extension.

Config.	TPPI	FPPI	Coverage	Time (ms)
$r = 0.05$	1.009	1.345	0.614	0.236
$r = 0$ (baseline)	0.966	0.871	0.594	0.233

The results demonstrate the contribution of the p_zr parameter in controlling border extension aggressiveness. Comparing the baseline ($p_zr = 0$) against the optimal configuration ($p_zr = 0.05$), we observe improvements in detection completeness: TPPI increases by **4.46%** (from 0.966 to 1.009) and coverage improves by **3.45%** (from 0.594 to 0.614). The TPPI of 1.009 approaches the ground-truth object density (TOPI = 1.034), indicating near-complete detection while still remaining below the theoretical maximum. However, this enhanced sensitivity comes at the cost of increased FPPI by **54.46%** (from 0.871 to 1.345). Processing time remains nearly identical (0.233 vs 0.236 ms), indicating that border extension aggressiveness affects detection quality rather than computational cost.

The choice of p_zr depends on the application’s priorities and tolerance for false positives. For applications prioritizing detection completeness — where missing an object is more costly than false alarms — the optimal value ($p_zr = 0.05$) provides measurable gains in detection rate and localization quality. The **4.46% increase in TPPI** and **3.45% improvement in coverage** justify the **54.46% increase in FPPI** when detection completeness is paramount.

Conversely, for applications where false alarms are costly — the baseline ($p_{zr} = 0$) offers better balance. $FPPI = 0.871$ represents a more manageable false alarm rate while still achieving strong detection performance ($TPPI = 0.966$) and good localization (coverage = 0.594). The modest differences between baseline and optimal suggest that $p_{zr} = 0$ already provides well-calibrated border extension for general-purpose drone detection scenarios.

The data reveals that the p_{zr} parameter provides a practical mechanism for tuning the detection sensitivity versus false alarm trade-off without algorithmic modification. Applications can select parameter values based on their specific cost function: baseline for balanced performance, or positive values (0.01–0.05) for maximum detection sensitivity at the expense of higher false positives.

J. OVEREXPOSURE ATTENUATION FACTOR

The $ovexp_k$ (Overexposure Attenuation Factor) parameter controls how the algorithm handles the highest histogram bin (intensity 255), which often contains overexposed pixels from bright sky regions, as described in Section III-B. The histogram value at bin 255 is multiplied by $ovexp_k$ before Fourier transform, effectively attenuating the contribution of overexposed regions to threshold selection. Values close to 0 strongly suppress overexposure effects, while values near 1.0 leave the histogram largely unmodified.

We systematically evaluated $ovexp_k$ values ranging from 0.01 to 0.99 to determine the optimal setting. Figure 10 illustrates how performance metrics vary across this parameter range.

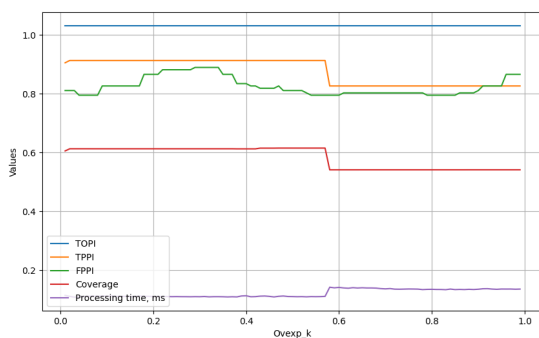


FIGURE 10. Performance metrics across different $ovexp_k$ values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from 0.01 to 0.99.

Table 8 compares the behavior with optimal attenuation versus minimal attenuation (baseline).

The results demonstrate that the $ovexp_k$ parameter has a modest but measurable impact on detection performance. The optimal range ($ovexp_k = 0.04$ – 0.08) achieves $FPPI = 0.871$, representing an **8.18%** reduction compared to the no-attenuation baseline ($ovexp_k = 0.99$, $FPPI = 0.948$). Critically, this $FPPI$ reduction comes with virtually no cost:

$TPPI$ remains identical at 0.966, and coverage decreases negligibly by only 0.45% (from 0.594 to 0.591).

The mechanism underlying this improvement is straightforward: overexposed pixels (intensity 255) in outdoor scenes typically correspond to very bright sky regions, clouds, or sun reflections—areas that rarely contain drone objects. By attenuating bin 255 in the histogram before Fourier transformation, the algorithm reduces the influence of these irrelevant bright regions on the threshold calculation. This prevents the threshold from being biased toward higher intensity values, which would otherwise cause the algorithm to miss darker foreground objects (drones) or incorrectly expand the threshold boundaries into bright background regions.

TABLE 8. Performance comparison: optimal vs. no overexposure attenuation.

Config.	TPPI	FPPI	Coverage	Time (ms)
$k = 0.04$ – 0.08 (optimal)	0.966	0.871	0.591	0.171–0.178
$k = 0.99$ (no atten.)	0.966	0.948	0.594	0.177

Performance remains relatively stable across a wide range of $ovexp_k$ values. $TPPI$ varies only from 0.957 to 0.966 (0.9% spread), while $FPPI$ ranges from 0.871 to 0.974 (11.8% spread) across the entire 0.01–0.99 parameter range. This stability suggests that our collected dataset has no sufficient overexposed images to have a visible impact on the overall algorithm performance.

The modest 8.18% $FPPI$ improvement observed in the original dataset requires important contextual interpretation. Modern surveillance cameras employ automatic exposure control that actively prevents overexposure by adjusting shutter speed, aperture, and gain. Consequently, in real-world footage from cameras with proper autoexposure, histogram bin 255 rarely becomes heavily populated, limiting the practical benefit of overexposure attenuation. The small improvement on the original dataset reflects this reality: with autoexposure functioning correctly, overexposure occurs infrequently, leaving little room for the $ovexp_k$ parameter to contribute.

To demonstrate the parameter's true contribution under conditions where overexposure is prevalent, we conducted an additional experiment with a doubled evaluation dataset that included the original images augmented with artificially overexposed versions of the same scenes. This simulates scenarios with very bright conditions, malfunctioning autoexposure, or cameras with manual exposure settings inappropriate for the lighting conditions. Figure 11 illustrates the performance comparison across the same $ovexp_k$ range on this overexposure-heavy dataset.

Table 9 compares the optimal attenuation against no attenuation ($ovexp_k = 0.99$) on the overexposure-heavy dataset.

On this overexposure-heavy dataset, the contribution of the $ovexp_k$ parameter becomes dramatically more

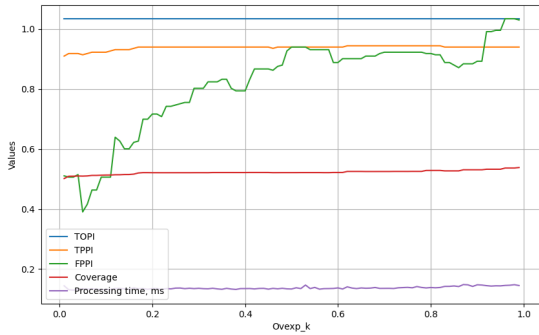


FIGURE 11. Performance metrics across different *ovexp_k* values on doubled dataset with additional overexposed images. The increased presence of overexposure clearly demonstrates the parameter's contribution when bin 255 is heavily populated.

TABLE 9. Performance on overexposure-heavy dataset: optimal vs. no attenuation.

Config.	TPPI	FPPI	Coverage	Time (ms)
$k = 0.05$ (optimal)	0.914	0.391	0.510	0.135
$k = 0.99$ (no atten.)	0.940	1.030	0.538	0.145

pronounced, achieving a **62.08%** FPPI reduction (from 1.030 to 0.391) compared to the 8.18% observed on the original dataset.

This stark difference validates the parameter's design rationale: when histogram bin 255 is heavily populated with overexposed pixels—precisely the condition the parameter targets—attenuation provides critical performance improvements. The comparison between datasets reveals an important practical consideration for deployment. In well-configured surveillance systems with functioning auto-exposure, the *ovexp_k* parameter provides modest but consistent improvement (8.18% FPPI reduction on the original dataset). However, in scenarios where overexposure cannot be avoided—extremely bright conditions, camera malfunction, manual exposure settings, or transitional lighting (e.g., cameras pointed toward sunrise/sunset)—the parameter's contribution becomes essential rather than supplementary, delivering a **62.08% FPPI reduction**. This characteristic makes *ovexp_k* a valuable robustness mechanism: it provides minimal overhead in normal conditions while offering substantial protection against performance degradation in challenging lighting scenarios.

The results demonstrate that the parameter's apparent modest impact on standard datasets should not be interpreted as limited utility, but rather as evidence of effective autoexposure in modern cameras. The doubled dataset experiment confirms that when the conditions the parameter is designed to handle actually occur, it delivers significant performance benefits, validating its inclusion as a core component of the algorithm.

K. LEFT BORDER ADJUSTMENT

The *lad* (Left Adjustment) parameter controls how the final low threshold is adjusted after identifying the left border of the background distribution, as described in Section III-C. Specifically, after determining the left border position through peak detection and boundary extension, the low threshold is set as: $\text{low} = \text{left} + \text{lad}$. Negative values shift the threshold leftward (toward darker intensities), expanding the region classified as background and making detection more conservative. Positive values shift the threshold rightward (toward brighter intensities), expanding the foreground region and increasing detection sensitivity.

We systematically evaluated *lad* values ranging from -30 to $+4$ to determine the contribution of this parameter to overall method performance. Figure 12 illustrates how performance metrics vary across this parameter range.

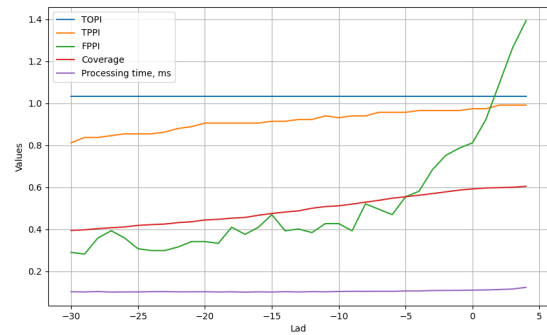


FIGURE 12. Performance metrics across different *lad* values. The plot shows TPPI, FPPI, Coverage, and Processing Time for values ranging from -30 to $+4$.

Table 10 compares the optimal *lad* configuration against the no-adjustment baseline ($\text{lad} = 0$).

TABLE 10. Performance comparison: optimal vs. no left border adjustment.

Config.	TPPI	FPPI	Coverage	Time (ms)
$\text{lad} = -9$ (optimal)	0.940	0.393	0.520	0.105
$\text{lad} = 0$ (no adj.)	0.974	0.812	0.593	0.111

The results demonstrate a critical contribution of the *lad* parameter in controlling the precision-recall trade-off. Comparing the optimal configuration ($\text{lad} = -9$) against no adjustment ($\text{lad} = 0$), we observe a dramatic **51.58%** FPPI reduction (from 0.812 to 0.393) at the cost of a modest **3.51%** TPPI decrease (from 0.974 to 0.940) and **12.20%** coverage reduction (from 0.593 to 0.520). Processing time improves slightly by **5.02%** (from 0.111 to 0.105 ms), indicating negligible computational overhead.

The mechanism underlying this trade-off reveals the parameter's strategic role. The left border position, determined by peak detection and boundary extension algorithms, identifies the approximate transition point between background and foreground intensities. However, this boundary is

rarely precise: histogram noise, gradual intensity transitions, and partial occlusion create ambiguity in the exact threshold location. Without adjustment ($lad = 0$), the threshold sits at this raw boundary position, capturing maximum foreground pixels but also including substantial background noise near the transition region.

By shifting the threshold leftward ($lad = -9$ moves it 9 intensity levels darker), the algorithm creates a safety margin that excludes this ambiguous boundary zone from foreground classification. Pixels near the background-foreground transition—those most likely to be misclassified as objects due to noise, shadows, or gradual intensity gradients—are now classified as background. This conservative approach dramatically reduces false positives: regions that barely fall below the original threshold (marginal candidates prone to being background noise) are filtered out, while true drone objects with distinctly dark intensities remain detected.

The exploration of positive lad values reveals the parameter's opposite effect. Setting $lad = +1$ to $+4$ shifts the threshold rightward into brighter intensities, expanding the foreground region. While TPPI increases slightly from 0.974 to 0.991 (+1.7%), FPPI explodes catastrophically: at $lad = +1$, FPPI rises to 0.923 (+13.7%); at $lad = +2$, it reaches 1.094 (+34.7%); at $lad = +4$, it peaks at 1.393 (+71.6%). This demonstrates that expanding the foreground threshold into brighter regions captures minimal additional true objects while incorporating massive amounts of background noise, rendering positive values impractical for deployment.

The trade-off manifests in three dimensions. First, TPPI decreases by 3.51% as some true objects near the threshold boundary are excluded. These are typically partial detections, highly faded objects, or objects in unusual lighting conditions where the drone's intensity approaches background levels. Second, coverage decreases by 12.20% because detected objects have tighter bounding boxes—the safety margin excludes peripheral pixels that were marginally below the original threshold, producing more conservative but less complete object boundaries. Third, FPPI decreases by 51.58%—the dominant effect—as spurious detections from background noise near the threshold are eliminated.

The choice of lad depends on the application's tolerance for false positives versus false negatives. For high-security surveillance where false alarms are extremely costly (triggering manual review, automated responses, or alarm fatigue), the optimal value ($lad = -9$) provides dramatic false positive reduction at minimal detection cost.

Conversely, for applications prioritizing detection completeness—where missing an object is more costly than false alarms—the baseline ($lad = 0$) maximizes TPPI (0.974) and coverage (0.593) while accepting higher false positive rates.

The data reveals that the lad parameter provides a powerful tuning mechanism for optimizing the detection system's operating point on the precision-recall curve. The 51.58% FPPI reduction achieved through optimal adjustment validates its inclusion as an essential

component for practical deployment, enabling systems to adapt to different operational requirements without algorithmic modification.

V. EXPERIMENTAL RESULTS

We evaluate the proposed method on 116 benchmark images for performance and for quality metrics, comparing against 12 existing thresholding methods: intermodes, isodata, li, max_entropy, mean, min_error, minimum, moments, otsu, percentile, triangle, and yen.

Although 116 images is small compared to modern learning-based benchmarks, it is sufficient for deterministic, training-free methods such as the thresholding algorithms considered here. Each method produces a single, parameterized output per image, and our ablation study explores thousands of parameter combinations across the entire dataset, yielding a dense sampling of the method's behavior under diverse background and lighting conditions. Nevertheless, extending the evaluation to larger, multi-camera datasets and continuous video sequences remains an important direction for future work.

A. PERFORMANCE COMPARISON

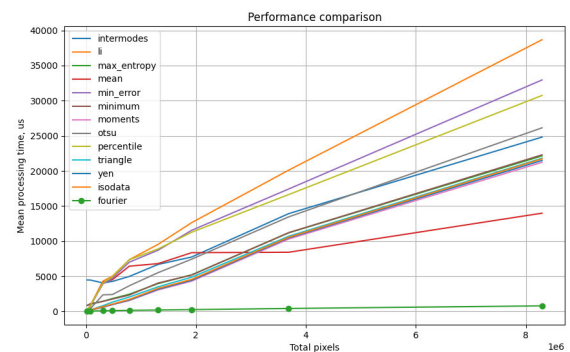


FIGURE 13. Average processing time versus image size (total pixels) for selected methods.

Processing times were measured over 100 runs per configuration, averaged across 5 experiments and all images. To ensure scientific validity and reproducibility, all benchmarks and tests were conducted on the same testing platform under identical conditions. For each method, multiple attempts (3 independent experiments) were performed, and the most performant results were selected to represent the optimal processing time achievable on the current testing platform under ideal conditions. This approach eliminates variability from system state fluctuations (background processes, thermal throttling, memory cache effects) and presents the true baseline performance capabilities of each method.

The testing platform specifications are as follows:

- **CPU/GPU:** Apple M3 Pro
- **RAM:** 18 GB
- **System:** MacBook Pro 16-inch

TABLE 11. Average processing times (ms) for different resolutions.

Res.	Prop.	Inter.	Iso.	Li	Max E.	Mean	Min E.	Min.	Mom.	Otsu	Perc.	Tri.	Yen
160x120	0.038	4.442	0.074	0.203	0.869	0.108	0.164	0.913	0.071	0.090	0.151	0.074	0.071
320x240	0.047	4.361	0.200	0.820	1.031	0.706	0.759	1.138	0.183	0.372	0.795	0.214	0.188
640x480	0.105	3.975	0.690	4.272	1.478	4.120	3.908	1.470	0.579	2.335	4.016	0.817	0.624
800x600	0.117	4.140	1.062	5.020	1.865	4.523	4.753	1.793	0.958	2.401	4.929	1.326	1.007
1024x768	0.158	4.834	1.730	7.344	2.473	6.410	6.978	2.403	1.554	3.644	7.314	2.058	1.627
1280x1024	0.210	6.560	3.285	9.500	3.994	6.718	8.670	4.062	3.073	5.472	8.843	3.525	3.160
1600x1200	0.269	7.659	4.620	12.562	5.273	8.309	11.569	5.242	4.354	7.416	11.244	4.900	4.457
2560x1440	0.436	13.809	10.637	20.090	11.240	8.400	17.349	11.339	10.319	13.544	16.599	10.767	10.490
3840x2160	0.817	24.733	21.777	38.679	22.238	13.858	32.915	22.370	21.268	26.107	30.732	21.931	21.586

The experiments utilized various image resolutions up to 3840×2160. Table 11 shows the average processing time (in milliseconds) for all methods across the resolutions.

Figure 13 plots the average processing time (in microseconds) versus the total number of pixels for the five selected methods.

The proposed method consistently outperforms the others in speed, with times ranging from 38 μ s at 160 × 120 to 817 μ s at 3840 × 2160. It scales exceptionally well, remaining under 1000 μ s even at ultra-high resolutions. Methods like Minimum, Yen, and Triangle show steeper increases with size, reaching over 21000 μ s at 3840 × 2160. This efficiency stems from the fixed-cost FFT on the 256-bin histogram combined with intelligent downsampling, making processing time largely independent of original image size.

It is important to emphasize that the proposed method operates in a fundamentally different computational regime than lightweight deep-learning detectors: our sub-millisecond CPU-only processing at 4K resolution is orders of magnitude faster than typical YOLO-like pipelines, which commonly report several milliseconds to tens of milliseconds per frame even on GPU-accelerated platforms. A systematic, hardware-matched comparison between the proposed Fourier-based thresholding approach and state-of-the-art lightweight deep detectors is therefore identified as promising future work rather than the primary focus of this study.

B. QUALITY COMPARISON

Quality is assessed using several metrics derived from detection performance. True Positives (TP) represent the number of correctly detected drone objects that match the ground-truth annotations. False Positives (FP) are the spurious detections where non-drone objects are incorrectly identified as drones. Misses refer to the ground-truth drone objects that the method failed to detect. Hits are the successfully detected drones, equivalent to TP.

The key metrics include:

- Precision: Precision = $\frac{TP}{TP+FP}$, measuring the proportion of detected objects that are actual drones.
- Recall: Recall = $\frac{TP}{TP+Misses}$, indicating the proportion of ground-truth drones that were successfully detected.

- Coverage: Coverage = IOU (Intersection over Union), where IOU is the average overlap ratio between the detected bounding rectangles and the ground-truth bounding boxes for the TP detections.

Table 12 presents the averages across the 116 benchmark images.

TABLE 12. Average quality metrics across the benchmark images.

Method	Precision	Recall	Coverage
Proposed	0.814	0.940	0.614
Minimum	0.729	0.844	0.414
Yen	0.662	0.840	0.546
Intermodes	0.649	0.839	0.560
Isodata	0.624	0.748	0.575
Max ent.	0.618	0.832	0.540
Otsu	0.350	0.442	0.358
Triangle	0.301	0.676	0.452
Li	0.137	0.245	0.188
Mean	0.103	0.182	0.133
Min err.	0.089	0.179	0.110
Percentile	0.068	0.142	0.091
Moments	0.034	0.034	0.013

The proposed method exhibits the highest precision (0.814) and recall (0.940), indicating superior accuracy in detecting and correctly identifying drones with fewer false positives and misses. With precision of 81.4% and recall of 94.0%, the method correctly identifies over 8 out of 10 detections as true drones while successfully detecting 94% of all ground-truth drones in the dataset. The coverage metric of 0.614 indicates good bounding box quality with an average 61.4% overlap with ground-truth annotations. While Isodata and Intermodes achieve slightly higher coverage (0.575 and 0.560 respectively), they do so at the cost of substantially lower precision (0.624 and 0.649), resulting in significantly more false positives. The proposed method offers the best balance of high precision, high recall, and competitive coverage, making it ideal for real-time drone detection applications.

Figure 14 provides a visual comparison of 6 methods, illustrating how the proposed method outperforms others in high-contrast scenes, such as those featuring bright white clouds compared to a small black drone in the center. This highlights the method's ability to achieve tighter and more accurate bounding boxes around the drone while minimizing

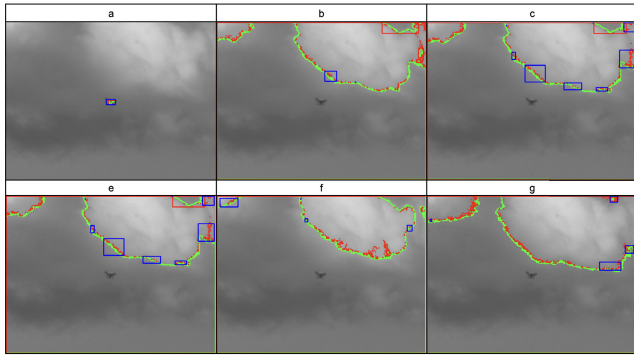


FIGURE 14. Qualitative comparison of detection results: (a) Proposed, (b) intermodes, (c) isodata, (d) max entropy, (e) minimum, (f) yen.

noise in the binary image. The intermodes, isodata, max entropy, minimum, and yen methods fail to detect the drone, instead only detecting the contour of the contrast clouds. In contrast, only the proposed method detects the drone with 43.6% object coverage, effectively identifying the drone in such conditions while the others primarily detect only the cloud contours.

C. LIMITATIONS AND APPLICABILITY

While the proposed method demonstrates robust performance on daytime imagery with sufficient lighting for low-ISO operation, certain limitations warrant discussion. The method is specifically designed and optimized for daytime detection scenarios where cameras operate with low ISO settings, producing images with minimal noise. This focus on daytime operation represents the primary use case for outdoor drone surveillance in well-lit conditions.

1) HIGH-ISO NOISY IMAGES

The proposed method can produce acceptable results on noisy high-ISO images when the target object is relatively large, as shown in Figure 15. However, detection performance degrades significantly for small objects under high-noise conditions. The histogram-based thresholding approach relies on identifying distinct intensity modes corresponding to background and foreground, which become obscured by noise in high-ISO imagery. For small objects, the noise amplitude may exceed the object's contribution to the histogram, preventing reliable threshold selection.

Processing of noisy and nighttime images is not covered in this paper, as we target daytime detection with lighting sufficient for low-ISO camera operation. For nighttime surveillance applications, thermal imaging cameras are recommended, as they produce low-noise images independent of visible light conditions.

2) THERMAL IMAGING COMPATIBILITY

The proposed method can be adapted for use with thermal camera imagery, as demonstrated in Figure 16.

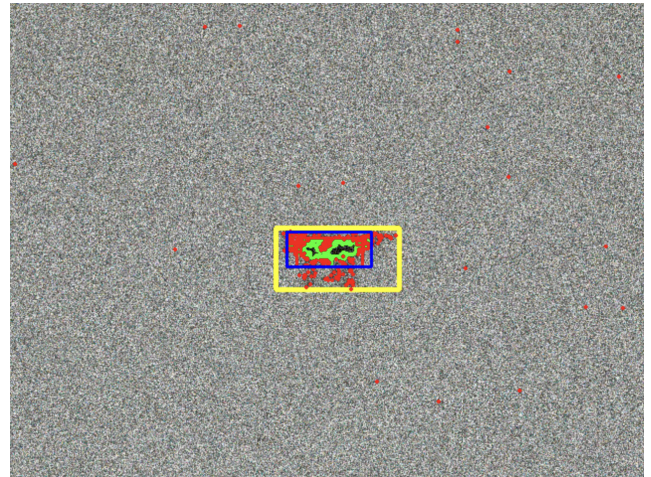


FIGURE 15. Example of drone detection on high-ISO noisy image. The method successfully detects the relatively large drone despite significant image noise, demonstrating some robustness to noise when object size is sufficient. However, small objects under similar noise conditions may not be reliably detected.

However, thermal images require preprocessing to invert the intensity values, as objects typically appear brighter than the background in thermal imagery (due to heat from motors, batteries, or body heat in the case of birds). By inverting the image during preprocessing, the algorithm's assumption of darker foreground objects against brighter backgrounds is maintained, enabling successful detection.



FIGURE 16. Example of drone detection on inverted thermal camera image. Three drones are successfully detected after color inversion preprocessing. In thermal imagery, objects appear warmer (brighter) than background due to heat from motors and batteries. Inverting the image maintains the method's core assumption of dark foreground against bright background.

Thermal imaging offers significant advantages for nighttime operation: images are inherently low-noise regardless of ambient lighting, and the thermal signature of drones (heat from electronics and motors) provides strong contrast against cooler backgrounds. The

method's compatibility with thermal imagery, demonstrated through simple color inversion, extends its applicability beyond daytime scenarios without requiring algorithmic modifications.

3) BRIGHT OBJECTS AGAINST DARK BACKGROUNDS

A fundamental limitation of the proposed method stems from its core design assumption: darker foreground objects (drones) against brighter backgrounds (sky). This assumption holds for the vast majority of daytime outdoor surveillance scenarios where drones appear as dark silhouettes against bright skies. However, the method cannot detect objects that are lighter than the background, such as white or light-colored drones flying against darker backgrounds (heavy clouds, dusk/dawn skies, or shadowed areas).

Figure 17 illustrates this limitation and its solution. The top image shows a white drone against a darker background that the algorithm fails to detect, as the intensity relationship violates the core assumption—the drone is brighter than the surrounding background, placing it in the intensity range the algorithm classifies as background rather than foreground. However, as shown in the bottom image, applying color inversion preprocessing reverses the intensity relationship: the white drone becomes dark, and the darker background becomes bright. With this simple transformation, the algorithm successfully detects the previously missed drone, as the inverted image now conforms to the expected dark-foreground, bright-background pattern.

This limitation is inherent to any contrast-based thresholding method designed for specific foreground-background intensity relationships. The practical impact is limited in typical surveillance deployments: most commercial and consumer drones are dark-colored (black, grey, dark blue), and daytime outdoor skies are typically brighter than objects, making the dark-foreground assumption valid for the overwhelming majority of cases. For scenarios where bright-colored drones are anticipated, the solution is straightforward: apply color inversion preprocessing to the input image before detection. This transformation adds negligible computational overhead (a single vectorized operation on pixel values) while enabling detection of bright objects.

Alternatively, a dual-processing pipeline could be implemented: run the detection algorithm on both the original image (to detect dark objects) and the inverted image (to detect bright objects), then merge the results. This approach provides comprehensive coverage of both dark and bright objects at the cost of doubling the processing time. Given the method's exceptional speed (0.817 ms at 4K resolution), a dual-pipeline approach would still achieve sub-2ms processing (over 500 fps throughput), remaining well within real-time constraints for most applications.

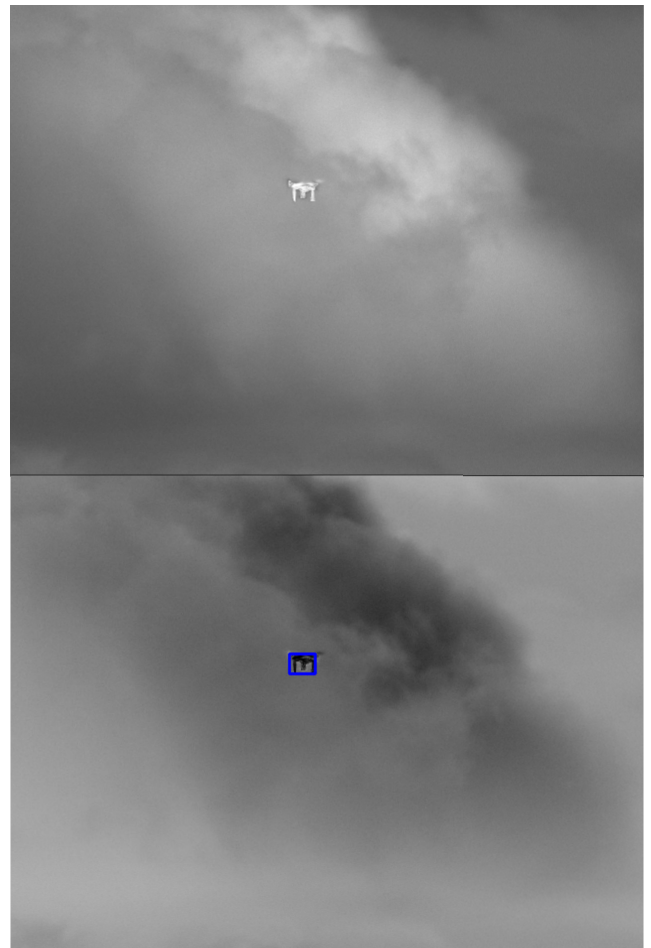


FIGURE 17. Detection of white drone against darker background. Top: Original image with undetected white drone—the algorithm fails because the drone is brighter than the background. Bottom: Same image with inverted colors—the white drone is now successfully detected.

In summary, the proposed method is optimized for daytime surveillance with low-ISO imaging but demonstrates adaptability to other modalities (large objects in noisy images, thermal imaging with inversion, bright objects with color inversion) under specific conditions. For comprehensive 24-hour surveillance, a hybrid approach combining visible-light cameras during daytime and thermal cameras for nighttime operation is recommended.

In addition to the above, several practical failure modes deserve explicit mention. First, extremely low-contrast scenes (e.g., heavy haze, fog, or strong glare) can produce histograms where the foreground-background separation becomes ambiguous, leading to either missed detections or increased false positives. Second, very small or distant drones that occupy only a few pixels may not contribute sufficient mass to the histogram to be distinguishable from noise, especially under motion blur. Third, cluttered backgrounds containing multiple dark, non-drone structures (trees, poles, buildings) can generate false alarms when their intensity falls into the same range as the drone silhouettes. While the method can be combined with simple priors (e.g., size

and aspect-ratio filtering, motion cues) to mitigate some of these effects, a more detailed treatment of such edge cases, including nighttime visible-light operation, is left for future work.

VI. CONCLUSION

This paper presents a novel Fourier-based adaptive thresholding method for real-time drone detection in outdoor surveillance scenarios. The key novelty of the proposed approach lies in its unique combination of histogram approximation via Fourier transform with multi-modal background peak detection, enabling robust threshold selection that adapts to complex outdoor lighting conditions while maintaining exceptional computational efficiency.

A. NOVEL CONTRIBUTIONS

The proposed method introduces several innovations that distinguish it from existing thresholding techniques:

Fourier-based histogram smoothing. Unlike conventional thresholding methods (Otsu, Triangle, Yen, Minimum, etc.) that operate directly on raw histogram data, the proposed method applies FFT to the intensity histogram and retains only low-frequency components (first 6–10 harmonics). This transforms the discrete histogram into a smooth continuous function that preserves dominant background modes while suppressing high-frequency noise from texture variations, ISO noise, and compression artifacts. Existing methods such as Otsu and Triangle are susceptible to local histogram irregularities that can cause threshold instability, whereas the Fourier approximation provides inherent robustness through low-pass filtering.

Multi-modal background detection. The method explicitly searches for multiple background peaks rather than assuming a simple bimodal distribution. Through iterative peak detection with configurable parameters (`nlp`, `npr`, `pli`), the algorithm identifies and excludes multiple background intensity modes—such as sky, clouds, ground, and structures—that commonly occur in outdoor scenes. This multi-peak capability surpasses single-threshold methods like Otsu, which assume a unimodal background and fail in complex outdoor environments with heterogeneous backgrounds.

Overexposure handling. The `ovexp_k` parameter provides targeted attenuation of histogram bin 255 before Fourier transformation, directly addressing the overexposure problem prevalent in outdoor imaging. Ablation studies demonstrate this parameter delivers a 62.08% FPPI reduction on overexposure-heavy datasets while maintaining minimal impact (8.18% FPPI reduction) on well-exposed imagery, indicating adaptive robustness to varying lighting conditions. Existing methods lack this explicit overexposure mitigation mechanism, leading to catastrophic failure when bright sky regions dominate the histogram.

Computational efficiency through decoupling. By performing downsampling before histogram computation and

applying FFT to the fixed-size 256-bin histogram, the method achieves processing time largely independent of input image resolution. The histogram analysis complexity is $O(256)$ regardless of whether the input is 640×480 or 3840×2160 , with only the initial downsampling step scaling with resolution. This architectural decision fundamentally differs from methods like Li, Isodata, and Mean, which operate directly on pixel data and exhibit quadratic or higher complexity growth with resolution.

B. QUANTITATIVE PERFORMANCE ADVANTAGES

Comprehensive benchmarking on 116 images across 13 thresholding methods reveals substantial quantitative advantages:

Quality superiority. Compared to the nearest quality competitor (Minimum method), the proposed approach achieves:

- **11.66% higher precision** (0.814 vs 0.729), indicating superior false positive control
- **11.37% higher recall** (0.940 vs 0.844), demonstrating better detection completeness
- **48.31% higher coverage** (0.614 vs 0.414), reflecting substantially improved bounding box accuracy

Speed superiority. At 3840×2160 resolution (4K video, representative of modern surveillance systems), the proposed method achieves:

- **27.4× speedup over Minimum** (0.817 ms vs 22.370 ms)
- **26.4× speedup over Yen** (0.817 ms vs 21.586 ms)
- **30.3× speedup over Intermodos** (0.817 ms vs 24.733 ms)

This computational efficiency enables real-time processing at 4K resolution: 0.817 ms per frame translates to 1,224 frames per second throughput on the test platform, far exceeding typical video framerates (30–60 fps). For comparison, the Minimum method at 22.370 ms per frame supports only 44.7 fps at 4K resolution, barely meeting real-time requirements for 30 fps video. At 2560×1440 resolution (common for modern surveillance cameras), the proposed method demonstrates 26.0× speedup over Minimum and 24.1× speedup over Yen, maintaining consistent performance advantages across resolutions.

C. PRACTICAL IMPLICATIONS AND FUTURE WORK

The combination of superior quality metrics and exceptional speed makes the proposed method particularly suitable for resource-constrained deployment scenarios: embedded systems, edge devices, and multi-camera surveillance installations where processing multiple high-resolution streams in real-time is required. The method's ability to maintain sub-millisecond processing even at 4K resolution enables deployment on modest hardware without requiring GPU acceleration, reducing system cost and power consumption.

The ablation study reveals several avenues for application-specific optimization. The `ovexp_k` parameter enables adaptation to camera autoexposure characteristics, while `lad` allows tuning precision-recall trade-offs based on false alarm costs. These tunable parameters suggest potential for automated parameter selection via reinforcement learning or Bayesian optimization in deployment, adapting the detector to site-specific conditions without manual intervention.

Limitations identified in this work—degraded performance on small objects in high-ISO noise and the need for color inversion with thermal imaging—suggest directions for future research.

The method's compatibility with thermal imaging, demonstrated through simple preprocessing, indicates potential for 24-hour surveillance deployments combining visible-light and thermal modalities.

In conclusion, the proposed Fourier-based adaptive thresholding method represents a significant advancement in real-time drone detection, offering substantial improvements in both quality (11.66% precision, 11.37% recall, 48.31% coverage) and speed (26–30× faster than competing methods) through novel applications of signal processing techniques to the thresholding problem. These quantitative advantages, combined with tunable parameters for deployment customization, position the method as a practical solution for real-world outdoor drone surveillance systems.

Complementary to our histogram-based frequency-domain processing, several recent works explore frequency-domain and learning-based normalization for visibility enhancement and dehazing, including wavelet-based physically guided normalization networks [15] and GAN-based approaches with fuzzy priors for robust image dehazing [16]. Such methods demonstrate that carefully designed frequency-domain or generative preprocessing can substantially improve contrast under haze and adverse weather [15], [16], [17]. Given that the proposed method already employs FFT-based processing on the histogram, integrating a compact wavelet-based normalization stage as a pre-processing step is a natural direction for future research, provided that the overall latency budget remains in the sub-millisecond range.

Beyond classical thresholding and fully supervised detection, there is a growing body of work on weakly supervised semantic segmentation and cross-domain representation learning, including image augmentation agents [18], adaptive patch-based contrast schemes [19], and contrastive prompt clustering [20]. Related research on sequential recommendation with large language models further illustrates how cross-domain signals and attention mechanisms can enhance localization and robustness in settings where only weak or indirect supervision is available [21]. These approaches operate in a fundamentally different regime than the hand-crafted, training-free method proposed here, but they

highlight complementary directions for integrating learned priors or cross-modal information into future hybrid drone-detection systems.

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